

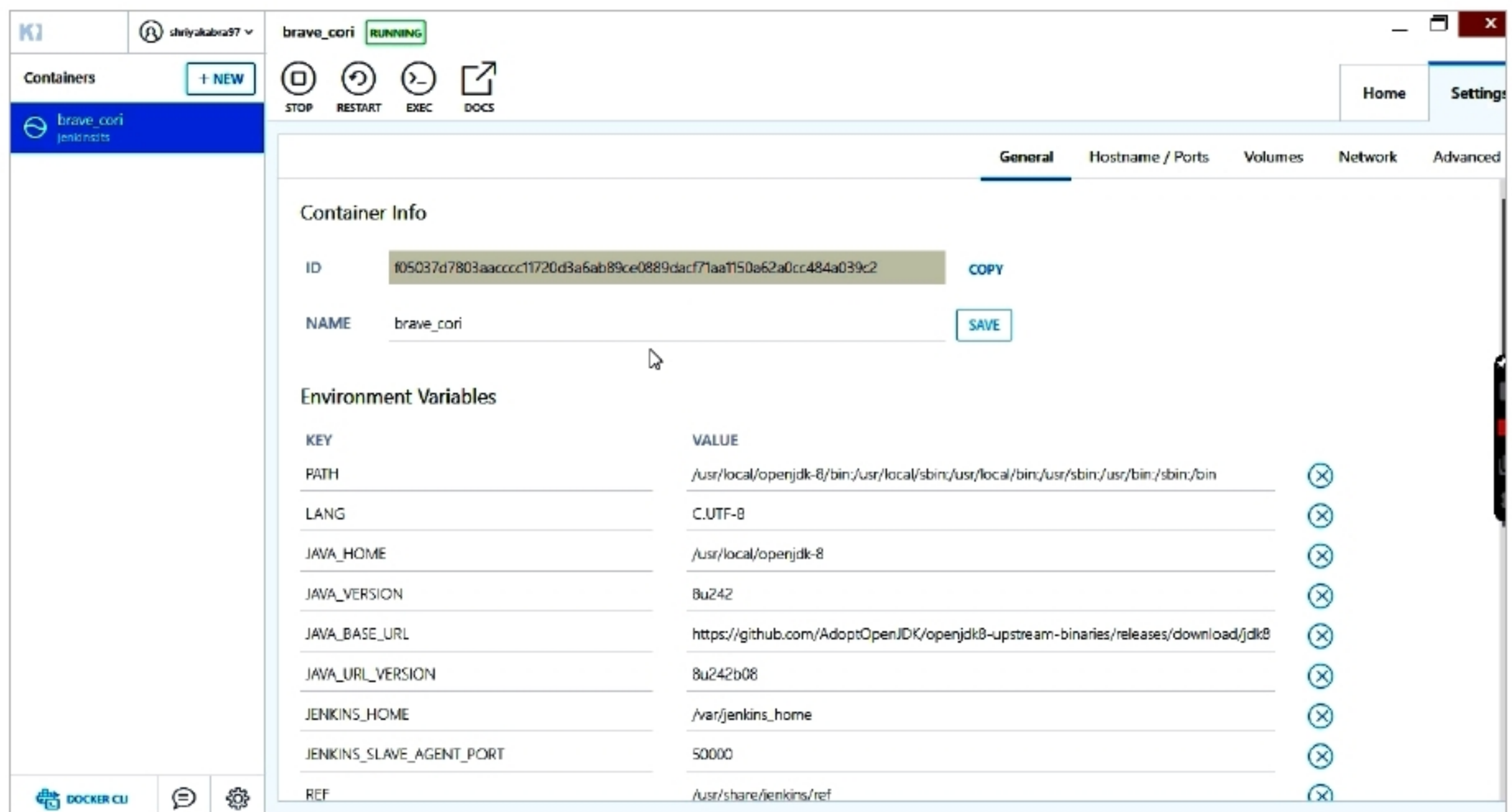


Figure 6: Properties of Jenkins container like container ID, container name and environment variables

- The tar file can be found in the *docker-images* directory. This tar file will be loaded on to the Docker Desktop using the *docker load* command, details of which will be covered later in the article.

If Docker Toolbox is great, why Docker Desktop?

Docker Toolbox has certain issues that Docker Desktop solves. A virtual machine is no longer a good option for Docker installation as new and better technology such as 'Windows Subsystem for Linux' is available and Docker Desktop uses this. Also, not all operations are possible with Kitematic. The option of pushing images is missing, and many tabs do not work as expected. Based on our experience, we recommend Docker Desktop rather than Docker Toolbox. If your system cannot support Docker Desktop, then Docker Toolbox is a good option for using Docker on Windows. The uninstallation procedure for Docker Toolbox is available at https://docs.docker.com/toolbox/toolbox_install_windows.

What is WSL?

Consider the task of running Linux distros on Windows machines. The obvious solution that comes to mind is running it in a virtual machine. Software like Oracle VirtualBox and VMware Workstation can help us with that. The software allocates some fixed computing resources to the virtual machine. Using a virtual machine puts heavy load on the host machine. This is because virtual machines use a static allocation of resources. Hence, when it is running, a chunk of the host machine's resources get blocked, which deteriorates its performance. Also software like Oracle VirtualBox and VMware Workstation are type-2 hypervisors, which means they access the host machine hardware indirectly through the host OS. This slows down the virtual machine too. Life would be much simpler if we could directly run our Linux distros without the involvement of such software.

If there were ways using which we could use basic Linux tools and applications directly on Windows, all

our worries would have gone away. Windows Subsystem for Linux (WSL) does exactly this. WSL allows us to run distros available on Microsoft Store on Windows machines. It allows processes running in Windows and Linux to communicate with one another. A client-server application stored on a Windows file system may have its GUI (client-side) running on Windows and its server-side running on Linux. Thus, WSL makes it possible to develop applications that can run on both Windows and Linux simultaneously.

WSL uses a hyper-optimised lightweight virtual machine, which is different from traditional virtual machines. It boots up in very little time and is completely managed by Windows itself. Thus, external software (like VirtualBox) is not needed.

WSL 1 vs WSL 2

WSL 2, released after WSL 1, is an improvement over the latter. The file system performance is much better in the second version. Also, it supports all the system calls supported by the Linux kernel. A detailed comparison between